

Sensor Technologies for Wastewater Collection Networks: Characteristics, Applications, and Limitations

Abstract

Wastewater collection networks can be monitored for water level, flow, water quality, gas conditions, and pipe condition. Each purpose requires a different sensor or inspection method. This review compares the main technologies used in sewer networks, with emphasis on their practical use, strengths, and limitations. Non-contact radar and ultrasonic sensors are widely applicable to level monitoring, although their performance depends on installation and environmental conditions. Flow estimates require both hydraulic context and suitable calibration. Water-quality probes, UV/Vis instruments, and fluorescence sensors can identify changes in wastewater composition, but fouling and matrix effects remain important. Hydrogen sulfide monitoring also requires a clear distinction between gas-phase and liquid-phase measurements. The review shows that sensor selection should begin with the operational question and the conditions at the intended site.

Keywords: wastewater collection; sewer monitoring; level sensing; flow sensing; water quality; optical spectroscopy; hydrogen sulfide; sensor placement.

1 Introduction

Sewer monitoring is used to detect surcharge and overflow risk, estimate hydraulic loading, identify unusual discharges, manage odour and corrosion, and assess pipe condition. These tasks do not produce the same type of information. A level sensor records hydraulic behaviour, a water-quality probe responds to changes in wastewater composition, and an inspection system examines the physical condition of the pipe. Telemetry then carries the measurements from the sewer to an operator or data platform.

The useful characteristics of a sensor depend on its location and purpose. Humidity, condensation, foam, solids, corrosion, limited access, radio coverage, and power availability can all influence performance. The following sections group sensor technologies by the practical question they answer and compare their applications and limitations.

2 Monitoring Questions and Sensor Categories

The literature is clearer when grouped by the question being answered rather than by a single list of hardware names.

Table 1: Question-led classification of sewer monitoring technologies.

Question	Examples	Information obtained	Main limitation
How full is the sewer?	Radar, ultrasonic, short-range LiDAR/ToF, float or tilt switches	Continuous level trends or discrete surcharge events	Humidity, condensation, foam, obstructions, fouling, or loss of trend information
How much is flowing?	Contactless ultrasonic flow, level plus geometry, pump drawdown or hydraulic models	Flow or velocity estimates and hydraulic loading	Installation geometry, flow direction, backwater, and calibration assumptions
What is in the wastewater?	pH, conductivity, temperature, turbidity, UV/Vis, fluorescence, H ₂ S	Chemical changes, illicit-discharge indicators, organic-load proxies, odour or corrosion indicators	Fouling, matrix effects, local calibration, and phase-specific measurement
Is the asset blocked or damaged?	Acoustic screening, CCTV, computer vision, fibre-optic or other condition methods	Structural or blockage information	Usually an inspection or diagnostic workflow rather than a continuous level time series
How does the reading get back?	LoRaWAN, cellular IoT, gateways, data platforms, soft sensors	Remote transmission and estimation of unmeasured states	Coverage, power, data quality, model dependence, and site-specific validation

Telemetry and soft sensors are enabling layers rather than physical hydraulic or water-quality sensors. Acoustic and CCTV methods are also distinct from fixed probes because they examine asset condition instead of producing a continuous level or chemical measurement.

3 Level and Depth Sensing

Continuous level data provide a time series of depth from which a utility may examine normal patterns, surcharge, and rate of change. A binary float or tilt switch provides an event threshold but cannot produce the same trend information. The choice therefore depends on whether the monitoring purpose is broad event notification, hydraulic diagnosis, or both.

3.1 Non-contact Radar

Radar level sensors use an electromagnetic signal reflected from the wastewater surface to measure the distance between the sensor and the water. Because the sensor is mounted above the flow, it avoids direct contact with wastewater. Its performance still depends on the selected instrument, mounting position, beam geometry, and site conditions.

Drenoyanis et al. [6] used non-contact radar for sewer level measurement and pumping-station drawdown tests. The study describes a portable drawdown device and a LoRaWAN-enabled prototype installed in the Shoalhaven Water network. A separate radio survey recorded communication distances of up to 20 km in an urban area and up to 35 km under line-of-sight conditions. These distances describe the LoRaWAN link, not the radar measurement range.

Radar was less affected by humidity and mist than ultrasonic sensing in the comparison reported by Drenoyanis et al. Steam, condensation, foam, and changes in air temperature were identified as potential sources of unstable ultrasonic readings. Radar nevertheless requires careful installation because an unsuitable mounting arrangement can produce false detections. Its higher cost can also limit the number of sites that can be instrumented.

The study presents an integrated 80 GHz short-range radar network, with approximately five devices around each pumping station, as a future development rather than a completed field deployment [6, pp. 28–29]. Instrument frequency, accuracy, service life, and power requirements therefore need to be considered for the particular radar unit selected.

3.2 Ultrasonic, LiDAR, and Time-of-Flight Sensors

An ultrasonic level sensor estimates distance from the travel time of an acoustic pulse reflected by the wastewater surface. It is a common non-contact option and can provide a continuous level record at relatively low power. Steam, mist, condensation, foam, and temperature-dependent changes in sound speed can cause unstable readings [6, pp. 3–4]. Nearby structures and an unsuitable mounting angle may also interfere with the acoustic path.

Utepov et al. [16] compared Sharp infrared, TF-Luna LiDAR, TOF400 laser, and JSN-SR04T ultrasonic sensors in a prototype designed to imitate a sewer well. In that arrangement, the JSN-SR04T measured to approximately ± 5 cm at distances up to 2 m, while the TF-Luna result was within approximately ± 8 cm at distances up to 5 m. The reported material cost of the complete prototype was about USD 170.07, excluding labour. The tests were conducted in an imitative setup, and the authors identified trials in different sewer systems as future work.

LiDAR and other optical time-of-flight sensors offer compact distance measurement and may suit locations with a clear optical path. Their exposed optical surfaces can be affected by condensation and deposits. Field use therefore depends on protection of the sensing window, suitable mounting geometry, and testing under the conditions found in the target sewer.

3.3 Binary Float and Tilt Sentinels

Float and tilt devices answer a simpler question: has wastewater crossed a set threshold? Their binary output is useful for distributed surcharge or overflow alerts, especially where a continuous level record is unnecessary. They cannot show the rate of rise or describe changes between the trigger and reset levels. Grease, debris, mechanical restriction, and the choice of mounting point can affect operation. Battery life, maintenance intervals, and cost vary between products and communication settings.

4 Flow and Velocity Measurement

For a partially full sewer, discharge depends on both the wetted cross-sectional area and the mean water velocity. A level sensor alone does not automatically provide a reliable flow measurement when the flow is non-uniform or the geometry and boundary conditions are uncertain.

4.1 Contactless Ultrasonic Flow

Alshami et al. [1] studied twelve sewer locations in Hong Kong using contactless ultrasonic flow and level sensors. The level–flow correlation coefficient was 0.36, and the average relative flow-rate error was 72.14% when the sensor results were compared with Manning-equation estimates. The result reflects the instruments, sites, and comparison method used in that study.

Possible causes of error included small pipe diameters, installation angle, inconsistent flow direction, water level, and the presence of multiple inlets or outlets. In practice, contactless flow sensors need site-specific installation and calibration, making flow measurement more demanding than level measurement alone.

4.2 Level-derived Flow and Manning Models

Manning-type level-to-flow estimates can be useful when their assumptions match the target sewer reach. In a blocked or backwatered reach, the water level may rise without a proportional increase in velocity or discharge. Reliable conversion therefore depends on hydraulic context, pipe geometry, calibration, and comparison with observed flow where possible. The 72.14% result reported by Alshami et al. illustrates this problem for one network and instrument configuration.

Drenoyanis et al. [6] used radar level measurements during pumping-station drawdown to estimate pump flow indirectly. This application differs from measuring flow in a partially full gravity sewer because it relies on the changing water volume within a defined wet well.

5 Water-Quality and Chemical Monitoring

Hydraulic sensors show how full the sewer is and how its level or flow changes. Water-quality sensors provide different information: they can indicate an unusual discharge, dilution, changing organic load, or conditions associated with odour and corrosion. These sensors are most useful at locations where a chemical change has a clear operational meaning, such as an industrial inflow, a pressure-sewer discharge, or an upstream and downstream comparison point.

5.1 Physicochemical Event Detection

Shi et al. [14] deployed 22 low-cost units measuring level, temperature, and electrical conductivity across an urban Melbourne catchment for more than one year. Changes in these measurements were used to identify dry-weather illicit discharges and short-duration events. Electrical conductivity and temperature are useful event indicators, although the cause of a change still has to be interpreted from its timing, location, and surrounding conditions.

Li et al. [8] evaluated a sewage-quality monitoring system measuring temperature, conductivity, turbidity, and pH. The design included wall-jet cleaning and self-diagnostic functions. It was evaluated within a three-year field programme, and one trial site operated continuously in raw sewage for up to 24 months without maintenance or failure. This performance was achieved by the particular system and cleaning arrangement used in the study.

5.2 UV/Vis and Fluorescence

Lepot et al. [7] reviewed UV/Vis methods for estimating total suspended solids and total or dissolved chemical oxygen demand. These instruments provide rapid, reagent-free measurements, but the relationship between the optical signal and the target parameter changes with the wastewater matrix. Local calibration is therefore important when an instrument is moved to a new site or used with a different wastewater composition.

Carstea et al. [3] found that fluorescence features such as tryptophan-like Peak T and humic-like Peak C can correlate with biochemical oxygen demand, chemical oxygen demand, or total organic carbon. The strength of these relationships varies with the sample matrix and measurement method. Overlapping fluorophores, inner-filter effects, turbidity, and fouling can also alter the signal. Fluorescence is therefore most useful as a rapid calibrated proxy rather than a direct measurement of these laboratory parameters.

Optical instruments can provide frequent measurements without chemical reagents, which makes them useful for detecting short events between manual samples. Their main practical challenges are fouling, cleaning,

calibration drift, and sensitivity to the local wastewater matrix. Cleaning may be provided by a wiper, flushing system, compressed air, or scheduled manual maintenance, depending on the instrument and installation.

Other water-quality technologies include nutrient ion-selective electrodes, dissolved-oxygen probes, and microbial fuel-cell biosensors. They can support specialised monitoring objectives, but they generally require more attention to calibration, fouling, selectivity, and maintenance than a simple level sensor. Their usefulness depends on whether the measured parameter leads to a clear operational response.

6 Hydrogen Sulfide and Corrosion Monitoring

Hydrogen sulfide is relevant to sewer odour, worker safety, and biogenic corrosion. Gas-phase sensors measure concentrations in the sewer headspace, while liquid-phase sensors measure dissolved sulfide in wastewater. The two measurements have different purposes, instruments, calibration needs, and suitable installation points.

6.1 Gas-Phase Sensors

Pacheco Fernández et al. [10] compared three commercial gas sensors at two sewer sites. OdaLog and MyData devices had stated ranges of 0–200 ppm, while the SulfiLogger range was 0–1000 ppm. The reported battery specifications were approximately eight months, three months at the stated transmission configuration, and two years for the respective devices. Measurement range, logging interval, and telemetry configuration therefore need to be considered together when selecting a gas sensor.

Agreement between the gas sensors was better at low concentrations, while deviations increased as the concentration rose. High humidity and device-management requirements were also important practical considerations. A suitable gas sensor therefore needs an adequate measurement range, resistance to sewer humidity, accessible calibration, and a logging interval that captures short concentration peaks.

6.2 Liquid-Phase Sensors

Despot et al. [4] studied online liquid-phase H_2S sensors, including UV/Vis and electrochemical instruments. Local calibration, interference from solids, and regular inspection affected their operation. Useful monitoring points included the end of a pressure sewer and locations downstream of chemical dosing, where dissolved-sulfide measurements can help assess septicity and treatment performance.

Pandey et al. [11] reviewed sensor-based approaches to H_2S monitoring and their role in understanding odour and corrosion. Long-term interpretation requires attention to sensor drift, the position of the sensor, ventilation, wastewater conditions, and the relationship between short concentration peaks and cumulative corrosion exposure.

7 Pipe Condition and Inspection Methods

Acoustic screening, CCTV, computer vision, and fibre-optic methods examine structural defects, obstructions, and changes in pipe condition [2,9,12,13,17]. Acoustic methods can screen long pipe sections and help prioritise further inspection. CCTV provides direct visual evidence but requires access, suitable flow conditions, and skilled interpretation. Computer vision can assist with consistent defect classification, although its performance depends on image quality and the data used to train the model. These methods complement continuous hydraulic monitoring rather than replacing it.

8 Fouling, Maintenance, Telemetry, and Power

Fouling, solids, condensation, calibration, and safe access influence the reliability and cost of sewer monitoring. Li et al. [8] demonstrated wall-jet cleaning for a multi-parameter water-quality system, while Despot et al. [4] found that liquid-phase H_2S sensors needed regular inspection and maintenance. Maintenance frequency depends on the sensor surface, wastewater characteristics, cleaning system, and ease of access.

Non-contact measurement avoids direct immersion and usually reduces fouling, but exposed surfaces and mounting hardware may still need inspection. Radar is sensitive to mounting and beam geometry, while ultrasonic and optical sensors can be affected by condensation or deposits. Contacting and submerged devices face greater exposure to solids, grease, and biofilm.

Drenoyanis et al. [6] provide an Australian LoRaWAN case study and regional signal survey. Radio performance below ground depends on cover material, burial depth, antenna position, gateway location, and the surrounding terrain. These factors need to be tested at the intended monitoring sites. Sikorski et al. [15] show that sensor coverage, battery capacity, sampling frequency, and cost are linked decisions. More frequent measurement and transmission improve temporal detail but increase energy use and data volume.

Operational planning should include installation access, cleaning, calibration, communications testing, battery replacement, and data-quality checks. These requirements can outweigh the purchase price when a sensor is difficult to reach or maintain.

9 Comparison of Sensor Characteristics

Table 2: Characteristics and applications of sewer monitoring technologies.

Technology	Main use	Useful characteristics	Limitations	Examples
Radar level	Continuous level and pump drawdown	Non-contact; relatively resistant to humidity and mist	Careful mounting and clear beam path; higher cost	Drenoyanis et al. [6]
Ultrasonic level	Continuous level and early surcharge detection	Non-contact; low-power options are available	Condensation, foam, temperature, and mounting conditions	Drenoyanis et al. [6]; Do et al. [5]
LiDAR/ToF	Short-range distance measurement	Compact and potentially low power	Optical fouling, condensation, and limited sewer field experience	Utegov et al. [16]
Float/tilt switch	Threshold alarm	Simple event notification	No continuous trend or rate-of-rise information	Product dependent
Contactless ultrasonic flow	Flow and level estimation	Avoids submerged installation	Sensitive to geometry, flow direction, water level, and calibration	Alshami et al. [1]
Physicochemical probes	Conductivity, temperature, pH, and turbidity	Detects changes in wastewater composition	Fouling, calibration drift, and interpretation of events	Shi et al. [14]; Li et al. [8]
UV/Vis and fluorescence	TSS and organic-load proxies	Rapid and reagent-free	Local calibration, matrix effects, and optical fouling	Lepot et al. [7]; Carstea et al. [3]
H ₂ S sensors	Headspace gas or dissolved sulfide	Supports odour, corrosion, and dosing control	Phase-specific calibration, humidity, solids, and maintenance	Pacheco Fernández et al. [10]; Despot et al. [4]
Telemetry and soft sensors	Remote data transfer and estimated states	Supports distributed monitoring and automated alerts	Coverage, power use, data quality, and model transfer	Drenoyanis et al. [6]; Sikorski et al. [15]

10 Sensor Selection Considerations

No single sensor provides complete information about a wastewater network. A practical monitoring programme may combine continuous level sensors at hydraulically important sites, simpler threshold devices for wider event coverage, targeted water-quality or H₂S sensors, and inspection methods for pipe condition. Selection should consider:

- monitoring purpose: choke or overflow warning, hydraulic calibration, source identification, compliance, H₂S, or asset condition;
- network topology, pipe geometry, existing sensors, critical assets, and historical events;
- access, confined-space risk, telemetry, power, fouling, calibration, maintenance, and redundancy;
- measurement range, accuracy, sampling interval, battery demand, and communication method;
- installation cost and the time required for cleaning, calibration, and replacement.

The monitoring purpose should be defined before choosing a technology. For example, a float switch may be adequate for a simple high-level alarm, while a radar or ultrasonic sensor is more useful when the rate and duration of surcharge are important. Water-quality and H₂S sensors are better suited to selected locations where changes can be linked to a source, corrosion risk, or treatment action. This purpose-led approach avoids installing complex instruments where their additional information would not change an operational decision.

11 Conclusion

Sewer monitoring is best approached through the operational question that needs to be answered. Radar, ultrasonic, LiDAR, and float devices provide different levels of information about surcharge and water level. Radar is attractive in humid environments, while ultrasonic and optical sensors may offer lower-cost alternatives where condensation, foam, and fouling can be controlled. Flow measurement is more demanding because pipe geometry, flow direction, backwater, and calibration all influence the result.

Water-quality and optical sensors add information that hydraulic measurements cannot provide, particularly for unusual discharges, organic-load changes, odour, and corrosion. Their usefulness depends on cleaning, calibration, and the ability to interpret the signal at a particular site. Gas-phase and liquid-phase H₂S monitoring should also be selected separately because they measure different parts of the sewer environment. Overall, the most useful sensor is the one that provides enough information for a defined operational response and can be maintained reliably at the intended location.

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